What is the definition of constrictive pericarditis?

How Do Patients With Constrictive Pericarditis Present? (LO #1)

Patients with constrictive pericarditis usually present with signs and symptoms relating to the impairment of the heart's diastolic filling; the impairment is caused by the heart's encasement by the inelastic pericardium. With impaired filling, cardiac output decreases, and patients may report dyspnea or fatigue on exertion. In addition, there might be signs of fluid overload, such as peripheral edema, ascites, and pleural effusions. This fluid overload occurs because blood returning to the heart cannot be accommodated by the stiff pericardium, leading to a backup of fluid into the systemic and pulmonary circulation.

congestion. These changes are primarily due to impairments in the ventricle filling process.

Decreased Cardiac Output. Pericardial fibrosis reduces the ease of filling (compliance) of the pericardium, preventing it from expanding as the heart receives incoming blood from the venous system.

As compliance decreases, the left ventricle end-diastolic volume also decreases. At this point, the pericardium starts exerting pressure on the heart (aka the "constriction"), causing elevation and equalization of the **end-diastolic pressures** in all four cardiac chambers. Because of this equalization, ventricular filling and cardiac output are greatly diminished, since small pressure differences between the chambers allow normal ventricular filling during diastole.

Another cause of decreased cardiac output is the enhancement of **ventricular interdependence**. In simple terms, this is the way that a change in the volume of one ventricle will alter the volume of the other ventricle. Normally, with increases in venous return (e.g., during inspiration when negative thoracic pressure develops), the volume of blood in the right ventricle (RV) causes the interventricular septum to bulge into the left ventricle, reducing the LV volume.

Normally, the LV size decreases only slightly. However, in patients with constrictive pericarditis, where the pericardium is fibrotic and inelastic, increases in right ventricular volume cannot be well accommodated, so the interventricular septum shifts toward the left, markedly decreasing the left ventricular filling. This results in decreased stroke volume and cardiac output during inspiration (Figure 1).

Decreased Venous Return. A second consequence of constrictive pericarditis is a decrease in venous return. Recall that inspiration normally causes a negative intrathoracic pressure, which is transmitted to the large thoracic veins and the heart (right atrium), causing their pressure to fall. This increases the pressure gradient between the extrathoracic veins and the right atrium, effectively creating a "thoracic pump" that pulls blood back into the heart.

However, in constrictive pericarditis, this lower intrathoracic pressure does not transmit to the heart because of the fibrosed pericardium. Venous return drops, causing venous congestion (e.g., JV distention). The decreased venous return will lower preload and also contribute to reduced cardiac output.

How Do We Diagnose Constrictive Pericarditis? (LO #3)

The diagnosis of constrictive pericarditis is challenging because its signs and symptoms are often nonspecific and can be falsely attributed to other causes. Hence, diagnosis requires a high degree of clinical suspicion. The findings from the patient's history and physical exam combined with those from echocardiography are usually sufficient to make the diagnosis.

History

As mentioned earlier with signs and symptoms of reduced cardiac output and fluid overload. The presence of these

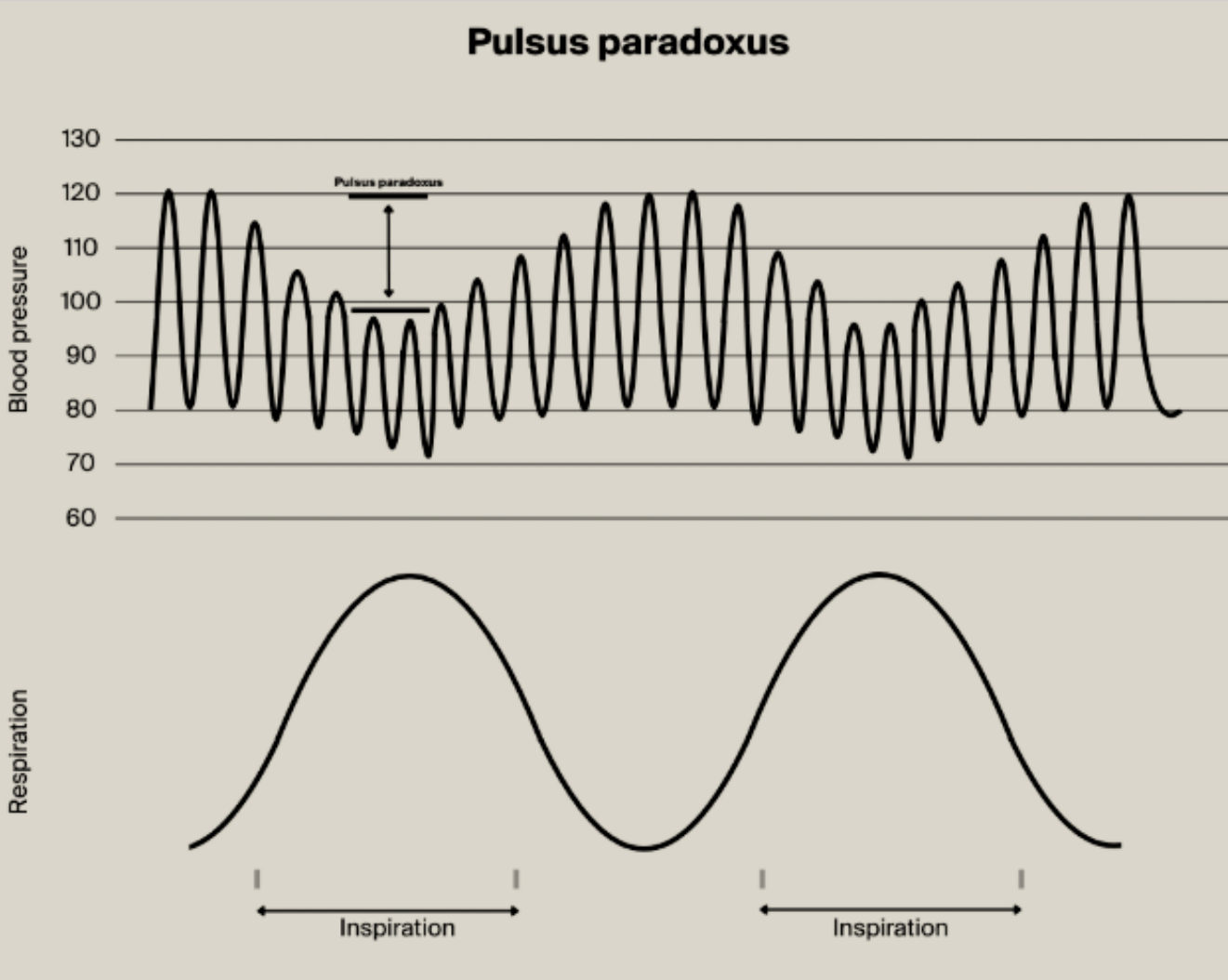

Figure 2. Pulsus Paradoxus. Note the drop in systolic blood pressure associated with inspiration.

A **pericardial knock** is a high-pitched early diastolic sound, corresponding to the abrupt cessation of ventricular filling as the ventricles stop expanding once the pericardial elastic limit is reached.

CLINICAL CORRELATION

Here is a link to hear a pericardial knock:

https://www.youtube

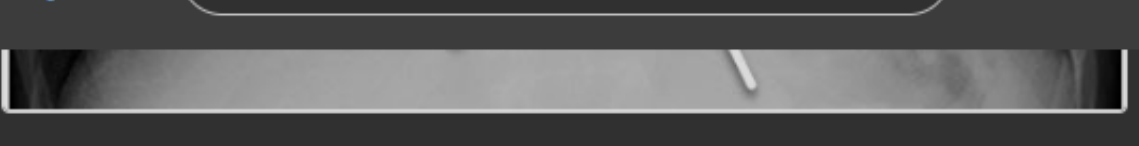

Figure 3. Calcifications in Constrictive Pericarditis. Note the curvilinear calcifications (white arrows) surrounding the heart in this patient with constrictive pericarditis.

Echocardiography is usually sufficient to confirm the diagnosis of constrictive pericarditis. It reveals the extent of pericardial fibrosis, as demonstrated by many parameters such as increased pericardial thickness or abnormal bulging of interventricular septum with inspiration. Sometimes, impaired diastolic filling in a constrictive pattern can be recognized.

Computed tomography (CT) or magnetic resonance imaging (MRI) may be used as alternatives to echocardiography in uncertain cases or when detailed anatomical visualization is required for surgical planning.

Invasive Hemodynamic Studies

In cases in which constrictive pericarditis is highly suspected, the gold standard diagnosis is invasive hemodynamic evaluation through cardiac catheterization. By using pressure catheters inside the right and left ventricles, the hemodynamic changes described above can be observed, such as elevation and equalization of diastolic blood pressures and exaggerated ventricular interdependence. There may also be elevated right atrial pressure or prominent y descent in the jugular venous tracing.

pericardiectomy, which is the removal of a portion or all of the pericardium (Figure 4). It is used for markedly symptomatic patients and often improves symptoms, but the operative risk is high (4%-12% operative mortality), so patients should be carefully selected.

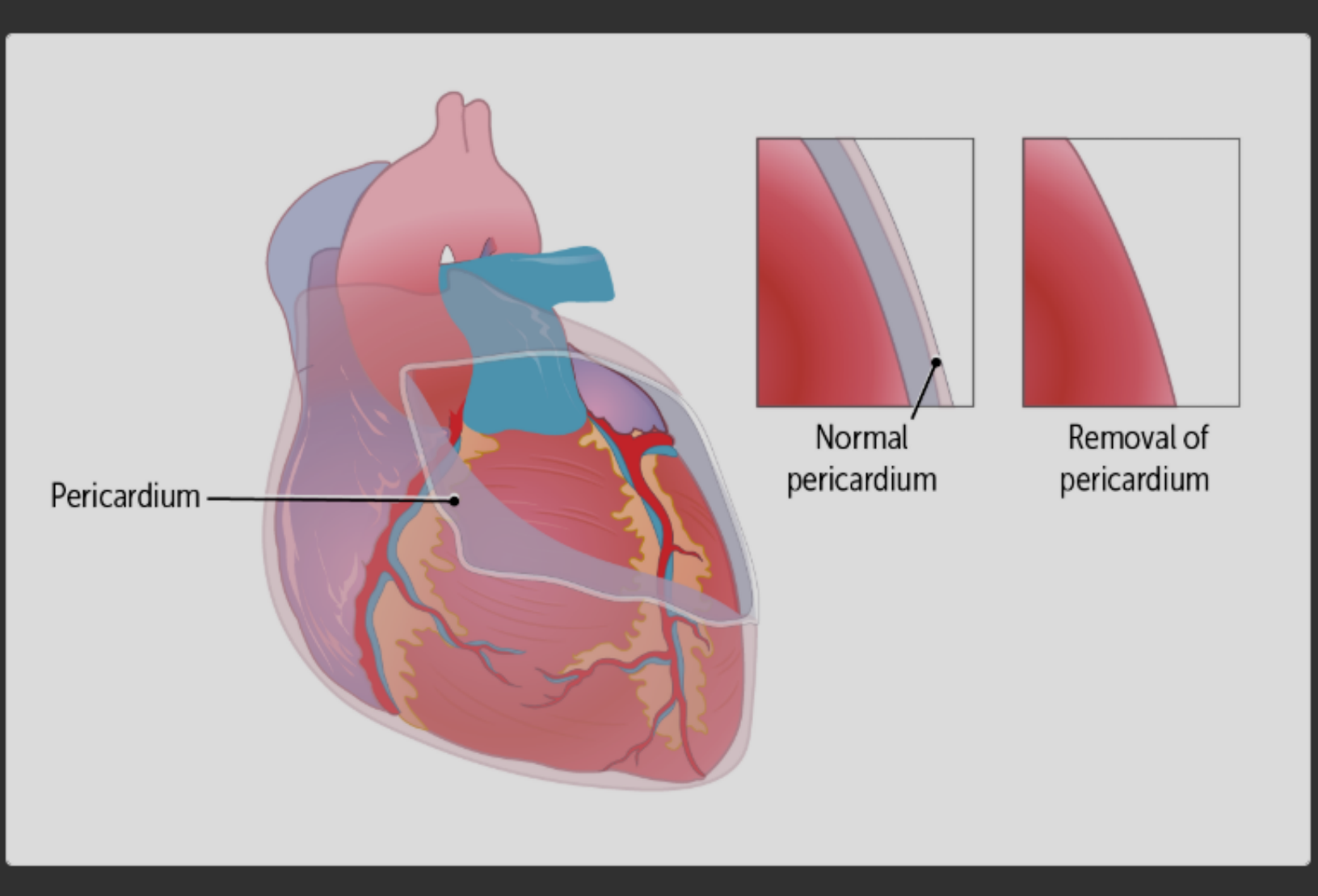

Figure 4. Pericardiectomy (Surgical Removal of the Pericardium).

formation of fibrous tissue throughout the pericardium.
- Constrictive pericarditis happens when the amount of blood allowed to fill the heart chambers is restricted by the fibrous tissue of the thickened pericardium.
- Patients present with signs and symptoms of reduced cardiac output, such as dyspnea on exertion, or fluid overload, such as peripheral edema.
- Constrictive pericarditis can be the end result of any pericardial disease.
- Constrictive pericarditis reduces cardiac output by impairing normal ventricular filling during diastole, equalizing end-diastolic pressures in all four cardiac chambers, and enhancing ventricular interdependence; thus, LV volume during inspiration is reduced.
- Physical exam findings in constrictive pericarditis include elevated jugular venous pressure, pulsus paradoxus, Kussmaul sign, and pericardial knock.
- ECG reveals nonspecific changes in constrictive pericarditis.
- Chest x-ray may reveal pericardial calcium deposits in constrictive pericarditis.
- The diagnostic confirmation of constrictive pericarditis is usually made through echocardiography, although CT or MRI are reasonable alternatives.
- Invasive hemodynamic evaluation is performed to confirm the diagnosis.
- Medical management for constrictive pericarditis includes anti-inflammatory drugs and diuretics.
- Pericardiectomy, a surgical treatment, may improve symptoms, but the operative risk is high.

Review Questions

1. Which of the following is least associated with constrictive pericarditis?

B. ○ Postcardiac surgery

C. ○ Postradiation therapy

D. ● Tuberculosis

E. ○ Viral infections

Hide Explanation

The correct answer is tuberculosis (D). Tuberculosis is the leading cause of constrictive pericarditis in developing countries, accounting for about 40%-90% of cases. In contrast, tuberculosis is an uncommon cause of constrictive pericarditis in developed countries, where it accounts for less than 5% of cases. In developed countries, constrictive endocarditis is often idiopathic or caused by various insults including connective tissue disorders (A), cardiac surgery (B), chest radiotherapy (C), and viral infections (E).

3. A 65-year-old male slowly develops constrictive pericarditis over several years after receiving thoracic radiation for a malignancy. He now reports increasing fatigue, dizziness, and dyspnea, and he has bilateral pleural effusions and peripheral edema on exam. Which of the following is the best next step in management?

A. ○ Colchicine

B. ● Furosemide

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of constrictive pericarditis.~~

02 ☑ ~~Explain the pathophysiology of constrictive pericarditis.~~

03 ☑ ~~Interpret clinical findings to properly diagnose constrictive pericarditis.~~

04 ☑ ~~Outline the management plan of constrictive pericarditis.~~

Objective Confidence: 4/4